



POLITECNICO
MILANO 1863

Apache Spark

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Rules

- Rename the `WebAccessAnalysisGroupXX.java` file replacing `XX` with the number of your group
- Write in the comment on top of the class your group number and the name of all group members
- Submit only a single java file with your solution
 - Submitted from the contact email provided in the group registration document

Assumptions

Two input datasets

1. accessLog

- Type: static, csv file
- Fields: UserID, PageURL, AccessTime, Duration

2. streamingData

- Type: dynamic, stream
- Fields: timestamp, value
- Each entry with value v indicates that someone visited the page with PageURL equal to v

Requirements

- For all queries: limit unnecessary recomputations as much as possible!
- For streaming queries: write the results on the console, showing only the results that changed since the last evaluation

Requirements

- Q1: compute the top 5 most popular pages from the accessLog, that is, the 5 pages that received the highest number of visualizations
- Q2: for each user u and for each popular page p (as computed in query Q1), compute the number of times user u visited page p
- Q3: for each popular page p (as computed in query Q1), compute the number of accesses to p within streamingData in a window of size 10 seconds, slide 4 seconds
- Q4: for each popular page p (as computed in query Q1), and for each window w (as computed in query Q3), compute the difference between the number of accesses to p within the accessLog and within w